

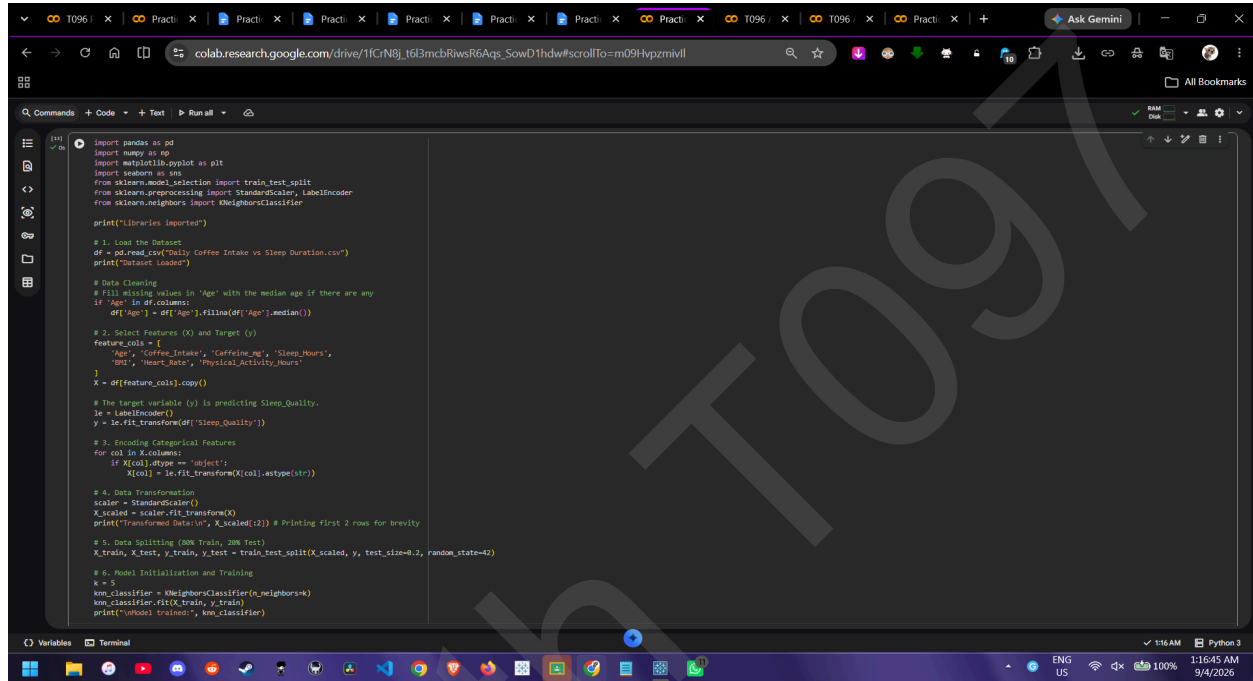
SHETH L.U.J & SIR M.V COLLEGE
Data Visualization with Power BI and Tableau

Practical No 8

Aim: K-Nearest Neighbors (K-NN)

- Implement the K-NN algorithm for classification or regression.
- Apply the K-NN algorithm to a given dataset and predict the class or value for test data.
- Evaluate the accuracy or error of the predictions and analyze the results.

Code:



```
[14]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler, LabelEncoder
from sklearn.neighbors import KNeighborsClassifier

print("Libraries Imported")

# 1. Load the Dataset
df = pd.read_csv("Daily Coffee Intake vs Sleep Duration.csv")
print("Dataset Loaded")

# Data Cleaning
# Fill missing values in 'Age' with the median age if there are any
if 'Age' in df.columns:
    df['Age'] = df['Age'].fillna(df['Age'].median())

# 2. Select Features (X) and Target (y)
feature_cols = [
    'Age', 'Coffee_Intake', 'Caffeine_mg', 'Sleep_Hours',
    'BMI', 'Heart_Rate', 'Physical_Activity_Hours'
]
X = df[feature_cols].copy()

# The target variable (y) is predicting Sleep_Quality.
le = LabelEncoder()
y = le.fit_transform(df['Sleep_Quality'])

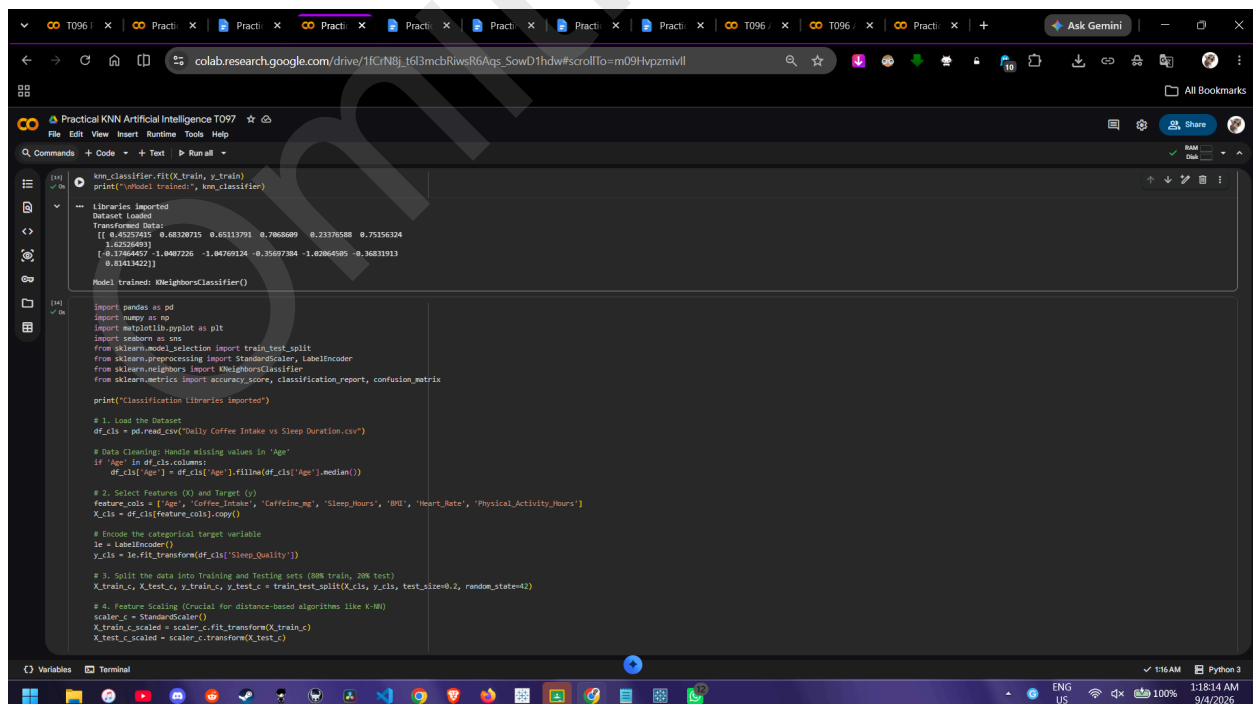
# 3. Encoding Categorical Features
for col in X.columns:
    if X[col].dtype == 'object':
        X[col] = le.fit_transform(X[col].astype(str))

# 4. Data Transformation
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
print("Transformed Data (n", X_scaled[2]) # Printing first 2 rows for brevity

# 5. Data Splitting (80% Train, 20% Test)
X_train, X_test, y_train, y_test = train_test_split(X_scaled, y, test_size=0.2, random_state=42)

# 6. Model Initialization and Training
k = 5
knn_classifier = KNeighborsClassifier(n_neighbors=k)
knn_classifier.fit(X_train, y_train)
print("Model trained:", knn_classifier)
```

Output:



```
[14]: knn_classifier.fit(X_train, y_train)
print("Model trained:", knn_classifier)

Libraries Imported
Dataset Loaded
Transformed Data:
[[ 0.45207415  0.68320715  0.65113791  0.78636099  0.23370588  0.75156324
  1.65236093]
 [-0.17664457 -1.0407726  -1.04769124 -0.35697384 -1.42064505 -0.30831913
  0.03413422]]
Model trained: KNeighborsClassifier()

[14]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler, LabelEncoder
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix

print("Classification Libraries Imported")

# 1. Load the Dataset
df_cls = pd.read_csv("Daily Coffee Intake vs Sleep Duration.csv")

# Data Cleaning: Handle missing values in 'Age'
if 'Age' in df_cls.columns:
    df_cls['Age'] = df_cls['Age'].fillna(df_cls['Age'].median())

# 2. Select Features (X) and Target (y)
feature_cols = ['Age', 'Coffee_Intake', 'Caffeine_mg', 'Sleep_Hours', 'BMI', 'Heart_Rate', 'Physical_Activity_Hours']
X_cls = df_cls[feature_cols].copy()

# Encode the categorical target variable
le = LabelEncoder()
y_cls = le.fit_transform(df_cls['Sleep_Quality'])

# 3. Split the data into Training and Testing sets (80% train, 20% test)
X_train_c, X_test_c, y_train_c, y_test_c = train_test_split(X_cls, y_cls, test_size=0.2, random_state=42)

# 4. Feature Scaling (Crucial for distance-based algorithms like K-NN)
scaler_c = StandardScaler()
X_train_scaled = scaler_c.fit_transform(X_train_c)
X_test_scaled = scaler_c.transform(X_test_c)
```

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Code:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler, LabelEncoder
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix

print("Classification Libraries Imported")

# 1. Load the Dataset
df_cis = pd.read_csv("Daily Coffee Intake vs Sleep Duration.csv")

# Data Cleaning: Handle missing values in 'Age'
if 'Age' in df_cis.columns:
    df_cis['Age'] = df_cis['Age'].fillna(df_cis['Age'].median())

# 2. Select Features (X) and Target (y)
feature_cols = ['Age', 'Coffee_Intake', 'Caffeine_mg', 'Sleep_Hours', 'HRV', 'Heart_Rate', 'Physical_Activity_Hours']
X_cis = df_cis[feature_cols].copy()

# Encode the categorical target variable
le = LabelEncoder()
y_cis = le.fit_transform(df_cis['Sleep_Quality'])

# 3. Split the data into Training and Testing sets (80% train, 20% test)
X_train_c, X_test_c, y_train_c, y_test_c = train_test_split(X_cis, y_cis, test_size=0.2, random_state=42)

# 4. Feature Scaling (Crucial for distance-based algorithms like K-NN)
scaler_c = StandardScaler()
X_train_c_scaled = scaler_c.fit_transform(X_train_c)
X_test_c_scaled = scaler_c.transform(X_test_c)

# 5. Initialize and Train the K-NN Classifier
k = 5
knn_classifier = KNeighborsClassifier(n_neighbors=k)
knn_classifier.fit(X_train_c_scaled, y_train_c)

# 6. Predict and Evaluate
y_pred_c = knn_classifier.predict(X_test_c_scaled)

accuracy = accuracy_score(y_test_c, y_pred_c)
print(f"--- K-NN Classification Results ---")
print(f"Accuracy: (accuracy * 100):{2f}%")
print(f"Classification Report:\n", classification_report(y_test_c, y_pred_c, target_names=le.classes_))
```

```
knn_classifier.fit(X_train_scaled, y_train_c)

# 6. Predict and Evaluate
y_pred_c = knn_classifier.predict(X_test_scaled)

accuracy = accuracy_score(y_test_c, y_pred_c)
print(f"--- K-NN Classification Results ---")
print(f"Accuracy: (accuracy * 100):{2f}%")
print(f"Classification Report:\n", classification_report(y_test_c, y_pred_c, target_names=le.classes_))

# 7. Plotting the Confusion Matrix
plt.figure(figsize=(10, 6))
cm = confusion_matrix(y_test_c, y_pred_c)
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues',
            xticklabels=le.classes_,
            yticklabels=le.classes_)
plt.title('K-NN Classification: Confusion Matrix')
plt.xlabel('Predicted Label')
plt.ylabel('True Label')
plt.show()
```

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Output:

